

Curriculum Vitae

Personal details

Name: Arjan Cornelissen
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Research interests

My primary research area is the field of quantum algorithms. I investigate the quantum query, time and space complexity of computational problems, by proving lower bounds on these quantities and constructing quantum algorithms that are efficient with regards to these measures.

Work experience

PostDoc at Cambridge University, Department of Computer Science (September 2026 —)
Supervisor: Tom Gur

PostDoc at Simons Institute, University of California, Berkeley
Supervisor: Umesh Vazirani (September 2024 — August 2026)

PostDoc at IRIF, Université de Paris Cité (January 2023 — August 2024)
Supervisors: Simon Apers and Frédéric Magniez.

PhD at QuSoft, University of Amsterdam (October 2018 — December 2022)
PhD thesis: *Quantum multivariate estimation and span program algorithms.*
Supervisor: Māris Ozols, Promotor: Ronald de Wolf.

Internship Quantum Algorithm Design at IBM Research (May 2022 — August 2022)
Supervisor: Kristan Temme.

Internship Mathematical Consulting at SIOUX Lime (September 2017 — November 2017)
Supervisor: Lense Swaenen.

Education

Master Applied Mathematics at Delft University of Technology
Average grade: 9.4/10 (*cum laude*). (September 2016 — September 2018)
Master thesis: *Quantum gradient estimation and its application to quantum reinforcement learning.*
Supervisors: Martijn Caspers and Ronald de Wolf.

Bachelor Applied Mathematics & Applied Physics at Delft University of Technology
Average grade: 9.3/10 (*cum laude*). (September 2013 — July 2016)

Bachelor thesis: *Quantum Computation: Shor's algorithm*.
Supervisors: Jan van Neerven and Miriam Blaauboer.

Publications

Statistics: 11 publications in conference proceedings, including 1 in STOC, and 2 in SODA. 7 presentations at top quantum conferences, namely 3 presentations at QIP and 4 at TQC. 1 journal publication in Quantum.

Subcube Stifling (June 2026)

Arjan Cornelissen, Nikhil S. Mande, Nithish Raja. [arXiv:2607.04850](#)

Quantum algorithms for path and cycle containment problems (May 2026)

Arjan Cornelissen, Amin Shiraz Gilani, Subhasree Patro. [arXiv:2605.09017](#)

In proceedings of the International Conference on Randomization and Computation (RANDOM 2026)

Quantum Search With Generalized Wildcards (November 2025)

Arjan Cornelissen, Nikhil S. Mande, Subhasree Patro, Nitish Raja, and Swagato Sanyal. [arXiv:2511.04669](#)

In proceedings of the Conference on Theory of Quantum Computation, Communication and Cryptography (TQC 2026).

Randomized and quantum approximate matrix multiplication (October 2025)

Simon Apers, Arjan Cornelissen, Samson Wang. [arXiv:2510.08509](#)

In proceedings of the 51st conference on Mathematical Foundations of Computer Science (MFCS 2026)

Quantum walks through generalized graph composition (September 2025)

Arjan Cornelissen. [arXiv:2510.04973](#)

Do you know what q-means? (July 2025)

Arjan Cornelissen, Joao F. Doriguello, Alessandro Luongo and Ewin Tang. [arXiv:2308.09701](#)

Highlighted talk at the 9th International Conference on Quantum Techniques in Machine Learning (QTML 2025).

Accepted to the Quantum Computing Theory in Practice (QCTiP 2026)

Quantum algorithms through graph composition (April 2025)

Arjan Cornelissen. [arXiv:2504.02115](#)

Accepted to the Quantum Computing Theory in Practice (QCTiP 2026)

Accepted to the Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2026).

How to compute the volume in low dimension? (March 2025)

Arjan Cornelissen, Simon Apers and Sander Gribling. [arXiv:2503.02207](#)

In proceedings of the International Colloquium on Automata, Languages, and Programming (ICALP 2025).

Quantum sabotage complexity (August 2024)

Arjan Cornelissen, Nikhil S. Mande and Subhasree Patro. [arXiv:2408.12595](#)

In proceedings of Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2024).

Quantum multivariate estimation and span program algorithms (February 2023)

Arjan Cornelissen. PhD Thesis at the University of Amsterdam.

Quantum policy gradient algorithms (December 2022)

Sofiene Jerbi, Arjan Cornelissen, Māris Ozols and Vedran Dunjko. [arXiv:2212.09328](#)

In proceedings of Theory of Quantum Computation, Communication and Cryptography (TQC 2023).

Quantum tomography using state-preparation unitaries (July 2022)

Joran van Apeldoorn, Arjan Cornelissen, András Gilyén and Giacomo Nannicini. [arXiv:2207.08800](#)

In proceedings of ACM-SIAM Symposium on Discrete Algorithms (SODA 2023).

Presented at the 26th Conference on Quantum Information Processing (QIP 2023).

A Sublinear-Time Quantum Algorithm for Approximating Partition Functions (July 2022)

Arjan Cornelissen, Yassine Hamoudi. [arXiv:2207.08643](#)

In proceedings of ACM-SIAM Symposium on Discrete Algorithms (SODA 2023).

Presented at the 26th Conference on Quantum Information Processing (QIP 2023).

Improved Quantum Query Upper Bounds Based on Classical Decision Trees (March 2022)

Arjan Cornelissen, Nikhil S. Mande and Subhasree Patro. [arXiv:2203.02968](#)

In proceedings of Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2022).

Presented at the 17th Conference on Theory of Quantum Computation, Communication and Cryptography (TQC 2022).

Published in Quantum.

Exact quantum query complexity of computing Hamming weight modulo powers of two and three (December 2021)

Arjan Cornelissen, Nikhil S. Mande, Māris Ozols and Ronald de Wolf. [arXiv:2112.14682](#)

Near-Optimal Quantum Algorithms for Multivariate Mean Estimation (November 2021)

Arjan Cornelissen, Yassine Hamoudi and Sofiene Jerbi. [arXiv:2111.09787](#)

Presented at the 25th Annual Conference on Quantum Information Processing (QIP 2022).

In Proceedings of the 54th Annual ACM Symposium on Theory of Computing (STOC 2022).

Quantum algorithms for multivariate Monte Carlo estimation (July 2021)

Arjan Cornelissen, Sofiene Jerbi. [arXiv:2107.03410](#)

Scalable Benchmarks for Gate-Based Quantum Computers (April 2021)

Arjan Cornelissen, Johannes Bausch and András Gilyén. [arXiv:2104.10698](#)

Span programs and quantum time complexity (May 2020)

Arjan Cornelissen, Stacey Jeffery, Māris Ozols and Alvaro Piedrafito. [arXiv:2005.01323](#)

In Proceedings of the 45th International Symposium on Mathematical Foundations of Computer Science (MFCS 2020).

Quantum gradient estimation of Gevrey functions (September 2019)

Arjan Cornelissen, [arXiv:1909.13528](#)

Teaching experience

Remote Internship Supervisor at QuSoft, University of Amsterdam (April — August 2026)

I will be supervising an internship project at the University of Amsterdam, on writing software to compute explicit quantum algorithms using the adversary bound and completely-bounded polynomials.

Teaching Assistant at Delft University of Technology (September 2014 — September 2018)

I taught several courses at the Bachelor and Master of Applied Mathematics, with subjects including calculus, mechanics and relativity, complex analysis and partial differential equations.

Teaching Assistant at Stichting Studiebegeleiding Leiden (September 2014 — August 2015)
I taught mathematics to high school students, preparing them for their final exam.

Freelance Writer at Malmberg (September 2014 — August 2016)
I assisted in the writing of a mathematics textbook for high school students by providing real-world demonstrations of the theory.

Awards

ASML Graduation Prize for Mathematics (November 2018)
Koninklijke Hollandse Maatschappij der Wetenschappen (KHMW)
Award for the best master's thesis in mathematics in the Netherlands.

Young Talent and Encouragement Award (November 2014)
Koninklijke Hollandse Maatschappij der Wetenschappen (KHMW)
Award for obtaining the highest average grade among the first-year students of Applied Physics.

Grants

Unitary Foundation Microgrant (2025)
Supervisor on a microgrant from the Unitary Foundation, with the aim to hire an intern to work on writing classical software to develop quantum algorithms.

Professional activities

Member of the Program Committee for QIP 2026 (October 2025)
I was involved in selecting the best submissions for the biggest theoretical quantum conference of the year.

Organizer of a reading group on the adversary bound (September 2023 — December 2023)
I hosted ten sessions in a reading group on the quantum adversary bound.

Organizer of the junior seminar at QuSoft (April 2019 — June 2022)
I organized the biweekly junior seminar, where the group's PhD students and PostDocs gave a talk.

CWI PhD Activity Committee (January 2020 – December 2021)
I was co-chair of this committee, that organizes social events for employees of the research institute CWI.

Other activities

Contributor to JTS and GEOS (December 2025 — present)
These are the default open-source libraries for computational geometry on geographic data.

Resident's committee of Collège Néerlandais (September 2023 – August 2024)
I was part of this committee that organizes social activities for the residents of the student housing in which I lived.

Google Hashcode Finalist (April 2021)
We ranked 22nd in this team-based programming competition out of over 9000 competing teams.

Google Code Jam Contestant***(May 2021 & 2022)***

I ranked 1752nd and 2287th in the second round of the Google Code Jam competition.

Benelux Algorithm and Programming Contest (BAPC)***(Fall 2017, 2019, 2020 & 2021)***

This is a programming competition for teams from the Netherlands, Belgium and Luxembourg.

North-Western European Regional Contest (NWERC)***(November 2017)***

We ranked 42nd out of 119 competing student teams in this team-based programming competition in Bath, UK.

International Mathematics Competition (IMC)***(August 2017, July 2018, August 2019)***

I earned a silver medal at this worldwide mathematics competition in Blagoevgrad, Bulgaria.

National Inter-University Mathematics Olympiad (LIMO)***(May 2016, 2017 & 2018)***

We ranked 12th, 11th and 6th in this mathematics competition for Dutch and Belgian university teams.

FIRST Tech Challenge (FTC)***(March 2012 & 2013)***

I participated in an international robotics competition for high school students.

Skills

Programming skills:

I am comfortable with these programming languages:

- Python 2/3
- Java, C/C++
- MATLAB/Octave
- HTML/CSS/Javascript/PHP
- TI BASIC/Zilog Z80 Assembly

Language skills:

- Fluent: Dutch, English
- High-school level: French, German

Interactive skills:

I enjoy giving presentations and explaining complicated ideas in a simple manner.